Bargaining Experience and Reciprocity

A fixed-stage eBay pilot with seller-level validation
and an exploratory measurement comparison

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Abstract

I investigate whether bargaining experience changes the strength of reciprocal concession-making, rather than only the average concession level. Using 30,419 actual four-offer eBay Best Offer threads from 4,520 sellers in a nonrandom public-data prefix, I estimate a fixed-stage experience-by-concession interaction. The primary novice/experienced contrast uses 27,016 threads and 3,004 seller clusters. Its interaction is 0.0127 (clustered SE 0.0587, 95% CI $[-0.1023, 0.1277]$, p=0.829); a seller-level split and planned checks do not establish weaker reciprocity among experienced sellers. A benchmark response slope is positive for remaining-gap fractions, 0.090, but negative for price movements on a common scale, -0.097. This exploratory sign disagreement recurs across the seller split. An algebraic derivation shows why these different parameters can have opposite signs. The proposed contribution is an experience-response extension and a transparent measurement diagnostic, not a causal effect, a refutation of the literature, or verified publication-level novelty.

Reading guide. The main text explains the design and interpretation. Tables report all planned and exploratory focal estimates. Four figures show validation, robustness, selection and measurement sensitivity. Appendices include every planned-model coefficient and the graph data. All numerical outputs are consolidated in `results/results.json`; the LaTeX source contains its figures and references inline.

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1 Research question and contribution

Online bargaining records make it possible to observe both prices and the offer sequence. I study the buyer's first counteroffer and the seller's following counteroffer at the same stage across threads. The original interest was bargaining on ride platforms; I use eBay because the public archive supplies actual offer histories. It contains no inDrive data.

The question is: *does experience change how strongly a seller responds to an opponent's concession?* Conceding less on average and being less responsive to a buyer's new concession are different empirical claims.

Backus, Blake, Larsen and Tadelis (2020) study reciprocity and experience-related concession levels in eBay data. Their appendix already considers alternative percentage-concession measures. Fong and Waisman (2025) use an instrumental-variable design to study delay. These findings motivate my empirical question; I do not reproduce those papers' exact samples or identification in this pilot.

1.1 What is an extension, and what is already established?

1. **Existing foundation:** reciprocal concession patterns, bargaining experience and their relation to concession levels have been studied before. Comparing bargaining rounds alone or checking an alternative scale alone would not be a new contribution.
2. **Proposed extension:** interact the opponent's concession with seller experience at one fixed stage. I distinguish slope heterogeneity from average toughness and examine the same pre-set contrast on disjoint seller halves.
3. **Additional exploratory diagnostic:** document and explain a measurement discrepancy within the completed pilot. The common-price check was planned; its seller-split follow-up followed the observed disagreement.

My targeted literature search did not establish that the exact interaction design had already been published. This is a candidate student-project extension; I do not claim certified global novelty. I retain the null primary result and report all planned tests.

2 A simple response model and testable hypothesis

Let $x \in [0,1]$ denote the buyer's concession and $q \in [0,1]$ the seller's following concession, each as a fraction of its relevant price gap. I use the following illustrative reduced-form behavioral loss

$$L(q; e, x) = \frac{c}{2}[q - m(e)]^2 + \frac{\rho(e)}{2}(q - x)^2, \qquad c > 0, \quad \rho(e) > 0. \tag{1}$$

Here $m(e)$ is a preferred concession without the matching motive, and $\rho(e)$ is the weight on matching the opponent's concession. This is an illustrative response model, not a full alternating-offers equilibrium.

Proposition 1. *For $m(e), x \in [0,1]$, the unique minimizer is*

$$q^*(e, x) = \frac{c\, m(e) + \rho(e) x}{c + \rho(e)}.$$

Its response slope is positive. Holding c fixed, an experience-related decline in ρ reduces that slope, whereas a decline in m can reduce concession levels without changing the slope.

Proof. The first-order condition is $c(q-m)+\rho(q-x)=0$ and the second derivative is $c+\rho>0$. The solution is a convex combination of m and x, so it is feasible. Differentiation gives

$$\frac{\partial q^*}{\partial x}=\frac{\rho(e)}{c+\rho(e)}>0, \qquad \frac{\partial^2 q^*}{\partial e\,\partial x}=\frac{c\rho'(e)}{[c+\rho(e)]^2}.$$

The intercept component can change through m while the slope stays fixed. □

My directional hypothesis recorded before inspecting sample outcomes was that experienced sellers have a weaker response slope. Observational data cannot separately identify c, m and ρ, or attribute a slope to a reciprocity motive rather than strategic response or selection.

3 Data and sample construction

3.1 Public source, period and bounded download

I use the NBER anonymized eBay bargaining archive. I retained exactly the first 67,108,864 compressed bytes (64 MiB) with a byte-range download. It produced 2,234,913 offer rows, 1,324,190 thread IDs and 21,315 sellers. Observed offer timestamps span June 2012–December 2013. The sample is a contiguous, seller-grouped prefix, not a random or representative draw. I discarded the final seller block because the range may end mid-history. The download manifest records the range, source and checksum.

3.2 History filters and the fixed stage

I remove identical analysis-field records, inconsistent party identifiers, conflicting timestamps, non-alternating sequences, nonpositive prices, missing initial experience and automatic acceptance/rejection histories. Pre-terminal offers must have the source countered status. Histories are chronologically ordered, contain at most six offers, and require each observable interior concession to lie between the preceding offers. These restrictions select a subset; they are not the original paper's exact cleaning procedure.

Table 1: Sequential sample construction

Filter / sample	Offer rows	Threads	Sellers
Raw prefix	2,234,913	1,324,190	21,315
Drop potentially truncated last seller	2,227,948	1,320,212	21,314
Deduplicate identical analysis-field records	2,227,877	1,320,212	21,314
Strict alternating human offer histories	1,363,663	928,563	18,614
Observed initial experience for both parties	1,315,115	900,435	18,554
Every observed interior concession bounded in [0,1]	1,304,538	897,890	18,542
Incoming qualifying buyer offers	–	70,167	–
Four-offer analysis sample	–	30,419	4,520
Primary experience contrast	–	27,016	3,004

The first six rows are sequential history filters. The last three rows select fixed-stage analytical samples. The raw prefix is nonrandom; the last seller block is removed because it may be truncated.

The fixed stage consists of four prices: B_1 (initial buyer offer), S_2 (first seller counteroffer), B_3 (first buyer counteroffer), and S_4 (second seller counteroffer). Define

$$x=\frac{B_3-B_1}{S_2-B_1}, \qquad q=\frac{S_2-S_4}{S_2-B_3}, \qquad d=\frac{S_2-B_1}{S_2}. \tag{2}$$

Qualifying histories have positive denominators and $x, q \in [0, 1]$. For illustration only, $B_1 = 60$, $S_2 = 100$, $B_3 = 80$, $S_4 = 90$ imply $x = q = 0.5$. This numerical example is not an observation used in estimation.

Table 2: Descriptive statistics by recorded seller experience

Experience	Threads	Sellers	Buyer mean	Seller mean	Median S_2
0-9	742	640	35.70	26.43	167.50
10-99	3,403	1,873	34.89	25.51	120.00
100+	26,274	2,385	35.63	23.18	60.00

Concession means are percentages of the relevant price gap; prices use the source price units. Some sellers cross experience thresholds over time, so seller counts need not sum to the unique total. The observed minimum experience count is 1.

3.3 Experience definition and seller split

The codebook describes experience as previous threads, while the paper's Table I discussion describes a count including the current thread. I use the supplied count unchanged and label thresholds as recorded experience. The observed minimum is 1; the pre-set novice bin is below 10 and the experienced bin is at least 100. Counts 10–99 remain in descriptive and common-slope analyses but are excluded from the primary contrast. Some sellers cross thresholds over time; group seller counts therefore need not sum to the unique total.

Before fitting, I split seller IDs by SHA-256: the integer represented by the first eight hexadecimal digits determines the half by parity. No seller appears in both halves. Buyers can appear in both halves, and the comparison is a seller-level stability check, not a fully independent population experiment.

4 Regression specifications and uncertainty

4.1 Primary interaction

For the fixed-stage qualifying continuers, I estimate

$$q_i = \alpha + \beta x_i + \theta E_i + \delta(x_i E_i) + X_i'\eta + \mu_{m(i)} + \varepsilon_i. \tag{3}$$

$E_i = 1$ for a recorded seller count of at least 100; the reference count is below 10. X_i contains $\log(1 + \text{buyer experience})$, $\log S_2$ and d before the buyer's concession. Calendar-month effects use the initial offer month. I implement ordinary least squares as weighted least squares with unit weights. For the equal-seller check, I use inverse within-sample seller thread counts. For the seller fixed-effects check, I demean within seller and absorb seller intercepts.

The novice slope is β, the experienced slope is $\beta + \delta$, and δ is the focal difference. Seller-clustered covariance allows correlated errors across a seller's threads. Intervals and two-sided p-values use the cluster-count t reference. The fixed-effects variance adjustment also counts absorbed seller intercepts. It does not solve omitted slope heterogeneity.

4.2 Benchmark and measurement follow-up

I estimate the benchmark using all experience categories, continuous $\log(1 + \text{seller experience})$, the buyer experience/price/gap controls, and month effects. In the measurement follow-up, I keep these controls and threads fixed but change the exposure/outcome definitions. It includes gap

fractions; both price movements divided by S_2; and percentages of each party's own preceding price. I do not claim causal identification or apply a multiplicity adjustment.

5 Primary results and validation

Table 3: Primary OLS interaction regressions

Variable	All	Discovery	Validation
Buyer concession (β)	0.0716	0.0337	0.1020
	(0.0539)	(0.0739)	(0.0788)
Experienced indicator (θ)	-0.0359	-0.0491	-0.0251
	(0.0218)	(0.0308)	(0.0308)
Buyer concession × experience (δ)	0.0127	0.0452	-0.0113
	(0.0587)	(0.0821)	(0.0833)
Log(1 + buyer experience)	0.0007	-0.0005	0.0019
	(0.0016)	(0.0022)	(0.0022)
Log first seller counteroffer (S_2)	0.0028	0.0001	0.0057
	(0.0028)	(0.0034)	(0.0043)
Gap before buyer concession	-0.0189	0.0121	-0.0504
	(0.0264)	(0.0392)	(0.0332)
Constant	0.2185	0.2314	0.2093
	(0.0270)	(0.0380)	(0.0385)
Month fixed effects	Yes	Yes	Yes
Threads	27,016	13,550	13,466
Seller clusters	3,004	1,487	1,517
Interaction p-value	0.829	0.582	0.893

Dependent variable: seller remaining-gap concession q. Parentheses contain seller-clustered standard errors. Reference experience count: below 10; experienced: at least 100. Counts 10–99 are excluded. Month reference: June 2012. No significance stars are used; focal p-values and intervals are reported explicitly.

I estimate an interaction of 0.0127 with clustered SE 0.0587, 95% CI $[-0.1023, 0.1277]$ and p=0.829. A 10 percentage point larger buyer concession therefore has an experienced-minus-novice response difference of +0.13 seller percentage points, with CI $[-1.02, 1.28]$. The interval does not support the proposed directional prediction, and it does not establish equivalence.

Discovery and validation estimates are +0.0452 and −0.0113, respectively; both intervals include zero. The experience result is not stable evidence of a stronger or weaker response.

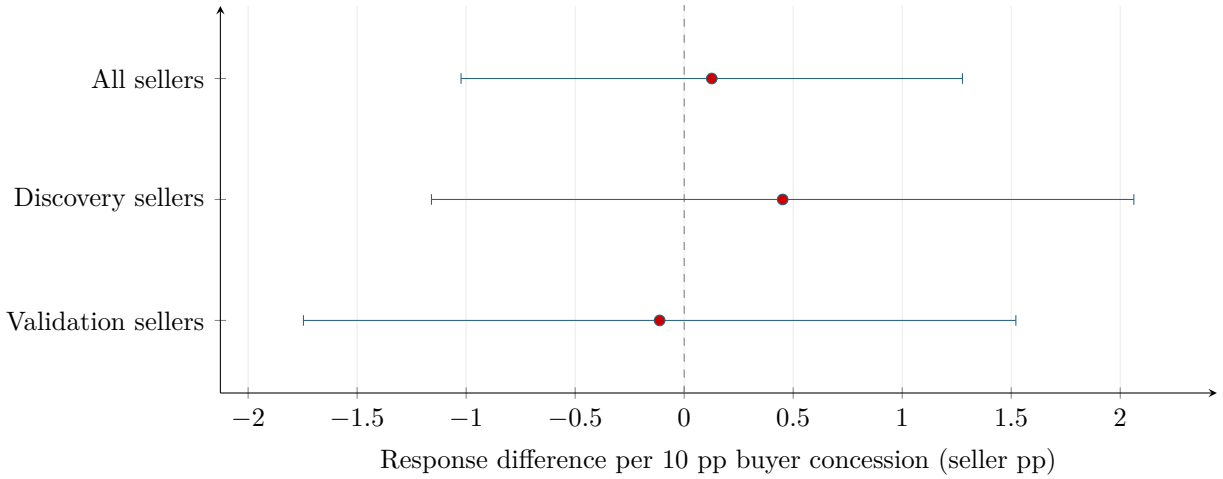


Figure 1: Primary experience interaction and seller-level validation. Points are adjusted slope differences expressed as the seller response in percentage points per 10 pp buyer concession. Whiskers are 95% seller-clustered intervals. Source: NBER eBay, June 2012–December 2013, nonrandom prefix.

In the all-experience benchmark, I estimate a positive gap-fraction response slope of 0.090 (SE 0.0176, CI $[0.055, 0.124]$). A 10 pp larger buyer concession is associated with a 0.90 pp larger seller concession. This resembles the published pattern but is neither an exact coefficient replication nor evidence that a norm caused it. Raw seller concession means are 26.43% for novices and 23.18% for experienced sellers. That level difference is descriptive; it does not establish an experience difference in response slopes.

6 Planned robustness and continuation selection

Table 4: All eleven planned interaction specifications

ID	Specification	N	δ	SE	95% CI	p
M1	All sellers	27,016	0.0127	0.0587	[-0.102, 0.128]	0.829
M2	Discovery	13,550	0.0452	0.0821	[-0.116, 0.206]	0.582
M3	Validation	13,466	-0.0113	0.0833	[-0.175, 0.152]	0.893
M4	Seller fixed effects	27,016	-0.0413	0.1116	[-0.260, 0.178]	0.711
M5	Experience 50+	28,462	0.0123	0.0583	[-0.102, 0.127]	0.833
M6	Experience 250+	24,395	0.0071	0.0595	[-0.110, 0.124]	0.905
M7	No messages	12,203	0.0792	0.1122	[-0.141, 0.299]	0.480
M8	Price 1–1,000	25,672	0.0318	0.0627	[-0.091, 0.155]	0.612
M9	Exclude midpoint / zero	15,918	-0.0292	0.0672	[-0.161, 0.103]	0.664
M10	Equal seller weights	27,016	0.0110	0.0606	[-0.108, 0.130]	0.856
M11	Common-price scale	27,016	0.0350	0.0286	[-0.021, 0.091]	0.221

M1–M10 use gap fractions. M11 scales both price movements by S_2 and estimates a different parameter. Confidence intervals are seller-clustered. None of these eleven planned interaction intervals excludes zero. The seller fixed-effects CR1 adjustment counts the absorbed seller intercepts conservatively.

Table 5: Group-specific slopes for every planned model

ID	Novice slope	SE	Experienced slope	SE	Novice N	Exp. N
M1	0.0716	0.0539	0.0843	0.0214	742	26,274
M2	0.0337	0.0739	0.0789	0.0337	351	13,199
M3	0.1020	0.0788	0.0908	0.0250	391	13,075
M4	0.1942	0.1097	0.1529	0.0182	742	26,274
M5	0.0716	0.0539	0.0839	0.0205	742	27,720
M6	0.0718	0.0540	0.0789	0.0232	742	23,653
M7	0.0777	0.1078	0.1569	0.0279	219	11,984
M8	0.0438	0.0579	0.0756	0.0224	649	25,023
M9	0.2555	0.0655	0.2263	0.0157	404	15,514
M10	0.0771	0.0564	0.0880	0.0232	742	26,274
M11	-0.1328	0.0287	-0.0978	0.0129	742	26,274

Model IDs are defined in Table 4. Experienced slope equals $\beta + \delta$, with its standard error calculated using the covariance between these terms. M11 has common-price rather than gap-fraction units.

The planned experience checks include seller fixed effects, 50+/250+ thresholds, no message threads, prices 1–1,000, exclusion of exact midpoints and zero concessions, equal seller weights and common-price scaling. All eleven planned interaction intervals include zero. Dropping messages leaves only 219 novice threads; precision depends on the small novice group, not only on the large total sample.

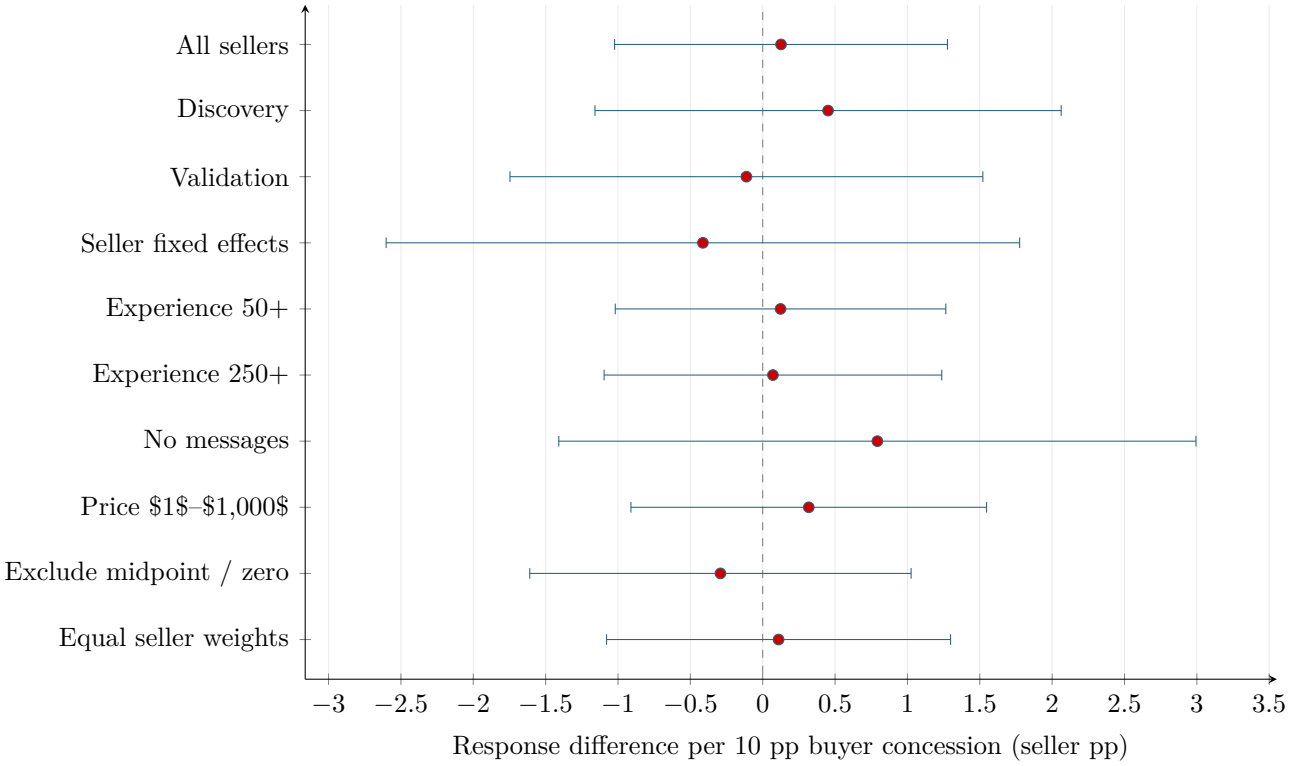


Figure 2: Planned gap-fraction interaction checks. Units are seller percentage points per 10 pp buyer concession; whiskers are 95% seller-clustered intervals. Common-price scaling is omitted from this figure because its units differ, but it is reported in Tables 4 and 5. Source: NBER eBay, June 2012–December 2013.

The secondary sample contains 70,167 qualifying incoming buyer counteroffers. Higher-concession offers are often accepted, so no next seller counteroffer is observed. The raw binned curves show this continuation selection; they do not correct it. The outcome is responsiveness conditional on a qualifying continued negotiation, not an unconditional response to every buyer offer.

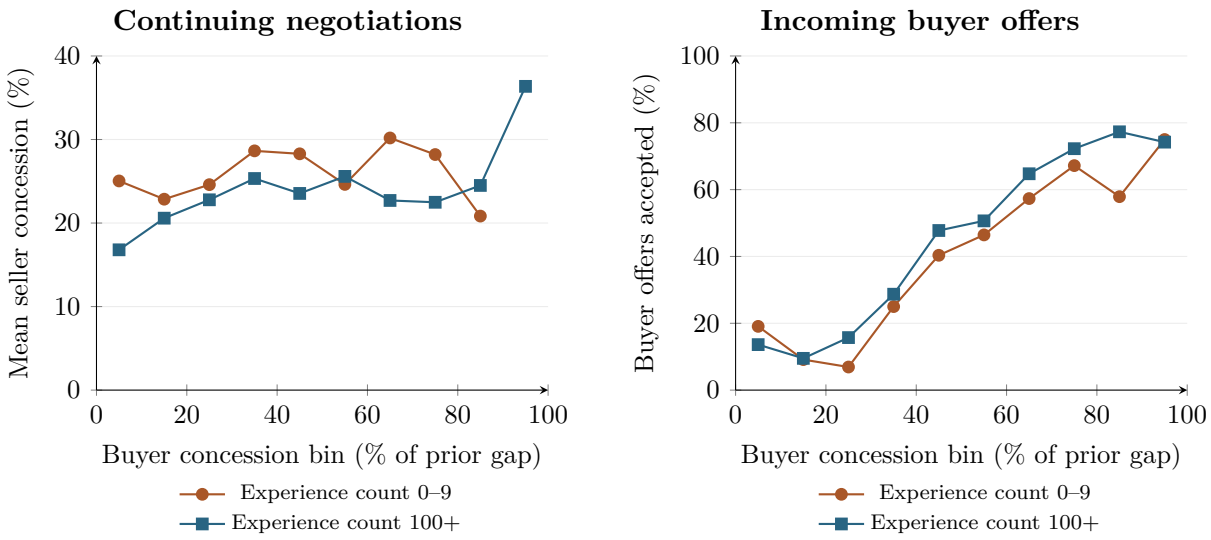


Figure 3: Raw concessions and response selection. Left: mean seller remaining-gap concession among four-offer continuers. Right: fraction of incoming buyer round-3 offers accepted. Points are 10 pp bins; groups use recorded experience. High-concession novice bins are sparse (only four continuing threads at 80–90% and none at 90–100%). Source: NBER eBay, June 2012–December 2013.

7 Exploratory measurement discrepancy

Using the same 30,419 threads and benchmark controls, I find a positive slope for gap fractions, $+0.090$ with CI $[0.055, 0.124]$, and a negative slope when both price movements are divided by S_2, -0.097 with CI $[-0.120, -0.075]$. The validation half reproduces the sign disagreement: $+0.096$ versus -0.088. These definitions estimate different parameters.

Table 6: Exploratory common-slope measurement comparisons

Measure	Sample	N	Slope	SE	95% CI	p
Remaining-gap fractions	all	30,419	0.0900	0.0176	[0.055, 0.124]	< 0.001
Price movements / S2	all	30,419	-0.0974	0.0116	[-0.120, -0.075]	< 0.001
Own preceding-price percentages	all	30,419	-0.0076	0.0041	[-0.016, 0.001]	0.067
Remaining-gap fractions	discovery	15,371	0.0859	0.0265	[0.034, 0.138]	0.001
Price movements / S2	discovery	15,371	-0.1072	0.0132	[-0.133, -0.081]	< 0.001
Own preceding-price percentages	discovery	15,371	-0.0051	0.0028	[-0.011, 0.000]	0.071
Remaining-gap fractions	validation	15,048	0.0963	0.0216	[0.054, 0.139]	< 0.001
Price movements / S2	validation	15,048	-0.0885	0.0178	[-0.123, -0.054]	< 0.001
Own preceding-price percentages	validation	15,048	-0.0164	0.0084	[-0.033, -0.000]	0.049

The published online appendix already checks own-prior-price percentage concessions (Table D2). Under that definition this pilot's pooled estimate is -0.0076 with CI $[-0.0156, 0.0005]$. The prefix, controls and lack of listing category/condition controls differ from the published specification. An exact replication is needed before arguing that a discrepancy is substantive. I therefore do not present alternative scaling itself as an unstudied idea.

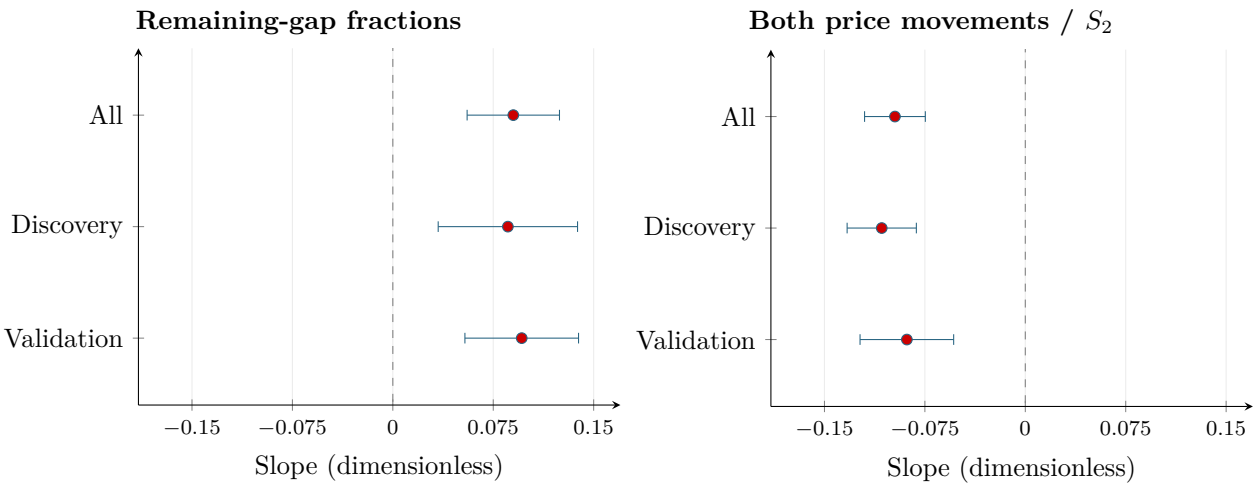


Figure 4: Exploratory measurement comparison on identical threads and controls. Points are regression slopes with 95% seller-clustered intervals. The two panels use different parameters; signs recur in the same seller halves. Source: NBER eBay, June 2012–December 2013, nonrandom prefix.

Table 7: Exploratory common-price experience interactions

Seller half	N	Novice slope	Exp. slope	Difference	p
discovery	13,550	-0.0881	-0.1109	-0.0228	0.515
validation	13,466	-0.1738	-0.0868	0.0870	0.044

Discovery difference CI: $[-0.0914, 0.0458]$; validation: $[0.0024, 0.1716]$. No multiplicity adjustment. A marginally significant validation estimate with a different-sign discovery estimate does not establish a stable experience effect.

I conducted the seller-split measurement comparison after the planned check. I had already used the validation set for the primary test. I added the own-price percentage diagnostic after locating the published appendix's definition. These analyses remain exploratory. In particular, the marginal validation p-value in Table 7 has no matching discovery result and no multiplicity correction.

7.1 Why opposite response signs can coexist

From Equation 2, define common-price movements $a = (B_3 - B_1)/S_2$ and $s = (S_2 - S_4)/S_2$. Then $a = dx$ and $s = d(1 - x)q$.

Proposition 2. *Holding $d > 0$ and other covariates fixed, suppose $\mathbb{E}[q \mid x, d, Z] = A + Bx$. Then*

$$\mathbb{E}[s \mid x, d, Z] = d(1 - x)(A + Bx), \qquad \frac{\partial \mathbb{E}[s \mid x, d, Z]}{\partial a} = B - A - 2Bx.$$

Consequently $B > 0$ need not imply a positive common-price response.

Proof. Substitute $s = d(1 - x)q$ into the conditional expectation. Differentiate with respect to x and divide by $\partial a/\partial x = d$. □

For example, $A = 0.2$ and $B = 0.1$ imply a positive gap-fraction slope but a negative common-price derivative for every $x \in [0, 1]$. These are illustrative parameters, not fitted estimates. The identity clarifies the measurement comparison; it does not identify motives or show that either measure is wrong.

8 Interpretation, limits and further research

I completed the proposed experience-response extension, but I did not find support for its directional hypothesis. A lower average concession can coexist with a similar estimated responsiveness. A null interaction is not a proof that experience never matters or that the groups behave identically.

The pilot is limited by nonrandom prefix sampling, selected continuation, observational experience, unobserved values/costs and missing listing category/condition controls. Seller fixed effects absorb mean differences, not endogenous experience, seller survival or unobserved response slopes. The measurement comparison is an exploratory discrepancy requiring exact replication of the published sample and definitions.

A stronger extension would use the full archive, join listing characteristics, study changes within sellers while separating experience from calendar time, and explicitly model acceptance versus counteroffering. An experiment randomizing bargaining practice would offer stronger identification. Larger samples improve precision but do not automatically solve these problems.

9 Reproducibility and project organization

My repository includes a single consolidated results file, processed empirical data, source provenance, shared regression code, internal design notes and a workflow for numerical verification and report compilation. The internal plan is not an externally registered pre-analysis plan. The repository is available at:

`https://github.com/nakshatra0904/bargaining-experience-reciprocity`

Reproduction: install `requirements.txt` with Python 3.12; run `scripts/run_analysis.py`, then `scripts/verify_results.py` and `scripts/build_report.py`. See the README for complete commands. The supplied compact CSV uses eight significant digits; the current reproduction agrees with the original primary interactions within 1.9×10^{-9}. Across regression and descriptive outputs, the maximum difference is 1.8×10^{-8}. Graph bins are recomputed from the supplied rounded sample, which can move observations at bin boundaries. The reference verification tolerance is 10^{-6}.

Compilation: this standalone source embeds graph coordinates and references. It requires no external figure or bibliography files. The report can be compiled in Overleaf or with a standard two-pass pdfLaTeX build. I compiled this PDF with hosted pdfLaTeX at TeXLive.net. Scientific output remains in this report; all machine-readable outputs remain in `results/results.json`.

References

[1] Backus, M., Blake, T., Larsen, B., and Tadelis, S. (2020). Sequential Bargaining in the Field: Evidence from Millions of Online Bargaining Interactions. *The Quarterly Journal of Economics*, 135(3), 1319–1361. doi:10.1093/qje/qjaa003.

[2] Backus, M., Blake, T., Larsen, B., and Tadelis, S. (2020). Online appendix, particularly Tables D1 and D2. Author-hosted appendix.

[3] Fong, J., and Waisman, C. (2025). The Effects of Delay in Bargaining: Evidence from eBay. *Management Science*. Published online 11 April 2025. doi:10.1287/mnsc.2023.01562.

[4] National Bureau of Economic Research. Public anonymized eBay bargaining archive associated with working paper 24306. Data and codebook. Source download and access recorded for this October 2026 pilot.

A Complete planned-model coefficients

Model identifiers M1–M11 are defined in Table 4. All coefficients, including month effects, are shown below; the seller fixed-effects model omits columns with zero within-seller variation. These are the actual fitted coefficient vectors, not hypothetical specifications.

Table 8: Complete coefficient vectors for the planned models

Model	Term	Coefficient	Clustered SE
M1	const	0.218520	0.027023
M1	buyer concession	0.071649	0.053939
M1	experienced	-0.035871	0.021775
M1	interaction	0.012654	0.058651
M1	log buyer experience	0.000694	0.001575
M1	log initial seller price	0.002842	0.002834
M1	gap before buyer concession	-0.018931	0.026434
M1	Month 2012-07	0.005550	0.009060
M1	Month 2012-08	0.002681	0.010213
M1	Month 2012-09	0.004391	0.010203
M1	Month 2012-10	0.002199	0.010556
M1	Month 2012-11	0.004403	0.010029
M1	Month 2012-12	0.012467	0.010193
M1	Month 2013-01	0.012370	0.009830
M1	Month 2013-02	0.015400	0.010441
M1	Month 2013-03	0.011609	0.009994
M1	Month 2013-04	0.012130	0.011148
M1	Month 2013-05	0.015344	0.010604
M1	Month 2013-06	0.013875	0.013301
M1	Month 2013-07	0.020721	0.015768
M1	Month 2013-08	0.021361	0.017170
M1	Month 2013-09	0.066399	0.018575
M1	Month 2013-10	0.041386	0.019003
M1	Month 2013-11	0.085451	0.023899
M1	Month 2013-12	0.082616	0.030660
M2	const	0.231398	0.038045
M2	buyer concession	0.033697	0.073910
M2	experienced	-0.049071	0.030767
M2	interaction	0.045176	0.082116
M2	log buyer experience	-0.000453	0.002170
M2	log initial seller price	0.000082	0.003370
M2	gap before buyer concession	0.012058	0.039240
M2	Month 2012-07	-0.002506	0.011969
M2	Month 2012-08	-0.003787	0.013375
M2	Month 2012-09	0.012139	0.012571
M2	Month 2012-10	0.010548	0.013566
M2	Month 2012-11	0.007968	0.013582
M2	Month 2012-12	0.023109	0.013297
M2	Month 2013-01	0.018897	0.012603
M2	Month 2013-02	0.016041	0.013218
M2	Month 2013-03	0.016160	0.012729

Model	Term	Coefficient	Clustered SE
M2	Month 2013-04	0.029575	0.015522
M2	Month 2013-05	0.028946	0.014606
M2	Month 2013-06	0.036054	0.017021
M2	Month 2013-07	0.024014	0.019584
M2	Month 2013-08	0.026866	0.021754
M2	Month 2013-09	0.081851	0.023594
M2	Month 2013-10	0.056547	0.025290
M2	Month 2013-11	0.093620	0.031062
M2	Month 2013-12	0.058601	0.044445
M3	const	0.209302	0.038457
M3	buyer concession	0.102007	0.078766
M3	experienced	-0.025080	0.030786
M3	interaction	-0.011254	0.083293
M3	log buyer experience	0.001881	0.002170
M3	log initial seller price	0.005745	0.004306
M3	gap before buyer concession	-0.050385	0.033216
M3	Month 2012-07	0.011775	0.013867
M3	Month 2012-08	0.006098	0.015018
M3	Month 2012-09	-0.004259	0.015630
M3	Month 2012-10	-0.007398	0.016402
M3	Month 2012-11	0.000178	0.014790
M3	Month 2012-12	0.000940	0.015644
M3	Month 2013-01	0.004894	0.015239
M3	Month 2013-02	0.013287	0.015960
M3	Month 2013-03	0.004882	0.015458
M3	Month 2013-04	-0.005884	0.015399
M3	Month 2013-05	0.000485	0.015034
M3	Month 2013-06	-0.010274	0.020297
M3	Month 2013-07	0.016761	0.025058
M3	Month 2013-08	0.016034	0.027963
M3	Month 2013-09	0.044296	0.030309
M3	Month 2013-10	0.024155	0.029480
M3	Month 2013-11	0.075762	0.037094
M3	Month 2013-12	0.102313	0.038363
M4	buyer concession	0.194188	0.109718
M4	experienced	0.050616	0.059412
M4	interaction	-0.041315	0.111631
M4	log buyer experience	-0.000625	0.000992
M4	log initial seller price	0.004504	0.002346
M4	gap before buyer concession	-0.095870	0.020875
M4	Month 2012-07	0.006580	0.009607
M4	Month 2012-08	0.008162	0.010395
M4	Month 2012-09	0.006067	0.009431
M4	Month 2012-10	-0.000913	0.010506
M4	Month 2012-11	0.008225	0.009529
M4	Month 2012-12	0.015443	0.009530
M4	Month 2013-01	0.008912	0.009425
M4	Month 2013-02	0.011231	0.009761

Model	Term	Coefficient	Clustered SE
M4	Month 2013-03	0.006879	0.009151
M4	Month 2013-04	0.009489	0.010051
M4	Month 2013-05	0.016138	0.009834
M4	Month 2013-06	0.016725	0.011304
M4	Month 2013-07	0.024406	0.013693
M4	Month 2013-08	0.003677	0.016473
M4	Month 2013-09	0.050341	0.016803
M4	Month 2013-10	0.040813	0.021026
M4	Month 2013-11	0.059331	0.022583
M4	Month 2013-12	0.068765	0.030600
M5	const	0.215360	0.026579
M5	buyer concession	0.071594	0.053936
M5	experienced	-0.034553	0.021651
M5	interaction	0.012317	0.058251
M5	log buyer experience	0.001039	0.001536
M5	log initial seller price	0.003056	0.002722
M5	gap before buyer concession	-0.021021	0.025203
M5	Month 2012-07	0.007785	0.008902
M5	Month 2012-08	0.003891	0.009964
M5	Month 2012-09	0.007067	0.010006
M5	Month 2012-10	0.006594	0.010298
M5	Month 2012-11	0.006161	0.009780
M5	Month 2012-12	0.014390	0.009911
M5	Month 2013-01	0.011627	0.009596
M5	Month 2013-02	0.016085	0.010143
M5	Month 2013-03	0.012013	0.009731
M5	Month 2013-04	0.014909	0.010753
M5	Month 2013-05	0.017005	0.010311
M5	Month 2013-06	0.016327	0.012928
M5	Month 2013-07	0.021159	0.015518
M5	Month 2013-08	0.023487	0.016776
M5	Month 2013-09	0.070543	0.018689
M5	Month 2013-10	0.042275	0.018754
M5	Month 2013-11	0.085174	0.023602
M5	Month 2013-12	0.080202	0.030281
M6	const	0.220184	0.027883
M6	buyer concession	0.071839	0.053962
M6	experienced	-0.037157	0.022052
M6	interaction	0.007064	0.059488
M6	log buyer experience	0.000596	0.001677
M6	log initial seller price	0.002656	0.003074
M6	gap before buyer concession	-0.016948	0.028737
M6	Month 2012-07	0.006794	0.009477
M6	Month 2012-08	0.002873	0.010752
M6	Month 2012-09	0.001419	0.010674
M6	Month 2012-10	-0.000309	0.011135
M6	Month 2012-11	0.003699	0.010464
M6	Month 2012-12	0.013531	0.010953

Model	Term	Coefficient	Clustered SE
M6	Month 2013-01	0.010820	0.010255
M6	Month 2013-02	0.013769	0.011144
M6	Month 2013-03	0.009056	0.010582
M6	Month 2013-04	0.010440	0.011883
M6	Month 2013-05	0.014873	0.011226
M6	Month 2013-06	0.010424	0.014054
M6	Month 2013-07	0.021625	0.016388
M6	Month 2013-08	0.021095	0.017690
M6	Month 2013-09	0.065474	0.019060
M6	Month 2013-10	0.038414	0.019383
M6	Month 2013-11	0.076419	0.024083
M6	Month 2013-12	0.089110	0.032088
M7	const	0.262244	0.046219
M7	buyer concession	0.077667	0.107849
M7	experienced	-0.049502	0.040978
M7	interaction	0.079242	0.112241
M7	log buyer experience	0.000495	0.002254
M7	log initial seller price	0.000174	0.004374
M7	gap before buyer concession	-0.076700	0.038232
M7	Month 2012-07	-0.013939	0.012999
M7	Month 2012-08	-0.015970	0.013936
M7	Month 2012-09	-0.012504	0.015687
M7	Month 2012-10	-0.004577	0.014068
M7	Month 2012-11	-0.008482	0.014404
M7	Month 2012-12	-0.002656	0.013188
M7	Month 2013-01	0.001610	0.013314
M7	Month 2013-02	0.010563	0.014106
M7	Month 2013-03	-0.004589	0.013142
M7	Month 2013-04	-0.003988	0.015293
M7	Month 2013-05	0.013723	0.013898
M7	Month 2013-06	0.012021	0.016686
M7	Month 2013-07	0.003485	0.020860
M7	Month 2013-08	-0.007645	0.020754
M7	Month 2013-09	0.026619	0.021916
M7	Month 2013-10	0.032087	0.025557
M7	Month 2013-11	0.092832	0.028505
M7	Month 2013-12	0.059226	0.040508
M8	const	0.217603	0.029301
M8	buyer concession	0.043808	0.057938
M8	experienced	-0.042076	0.023813
M8	interaction	0.031834	0.062675
M8	log buyer experience	0.000377	0.001619
M8	log initial seller price	0.004918	0.003476
M8	gap before buyer concession	-0.013140	0.027552
M8	Month 2012-07	0.007051	0.009298
M8	Month 2012-08	0.004911	0.010481
M8	Month 2012-09	0.006290	0.010541
M8	Month 2012-10	0.003235	0.010831

Model	Term	Coefficient	Clustered SE
M8	Month 2012-11	0.006740	0.010341
M8	Month 2012-12	0.014183	0.010472
M8	Month 2013-01	0.014247	0.010116
M8	Month 2013-02	0.016221	0.010782
M8	Month 2013-03	0.014664	0.010248
M8	Month 2013-04	0.014062	0.011458
M8	Month 2013-05	0.018054	0.010872
M8	Month 2013-06	0.015696	0.013421
M8	Month 2013-07	0.018938	0.016323
M8	Month 2013-08	0.023461	0.017564
M8	Month 2013-09	0.062820	0.019129
M8	Month 2013-10	0.048128	0.018893
M8	Month 2013-11	0.085133	0.023732
M8	Month 2013-12	0.082860	0.031807
M9	const	0.342583	0.032594
M9	buyer concession	0.255538	0.065450
M9	experienced	-0.025659	0.023777
M9	interaction	-0.029207	0.067191
M9	log buyer experience	0.003850	0.001501
M9	log initial seller price	-0.008183	0.003082
M9	gap before buyer concession	-0.176035	0.019913
M9	Month 2012-07	-0.017038	0.011312
M9	Month 2012-08	-0.020287	0.010296
M9	Month 2012-09	-0.017112	0.011540
M9	Month 2012-10	-0.020680	0.012717
M9	Month 2012-11	-0.010096	0.012186
M9	Month 2012-12	-0.006163	0.012040
M9	Month 2013-01	-0.003877	0.011745
M9	Month 2013-02	-0.006698	0.012028
M9	Month 2013-03	-0.007432	0.012761
M9	Month 2013-04	-0.010803	0.012922
M9	Month 2013-05	0.003626	0.013185
M9	Month 2013-06	-0.007178	0.015705
M9	Month 2013-07	-0.016649	0.016791
M9	Month 2013-08	-0.005042	0.019458
M9	Month 2013-09	0.018859	0.021516
M9	Month 2013-10	0.008595	0.022586
M9	Month 2013-11	0.036891	0.029395
M9	Month 2013-12	0.017465	0.029023
M10	const	0.181308	0.031748
M10	buyer concession	0.077058	0.056361
M10	experienced	0.001685	0.022894
M10	interaction	0.010970	0.060619
M10	log buyer experience	0.003063	0.002053
M10	log initial seller price	0.003212	0.002332
M10	gap before buyer concession	0.033795	0.022589
M10	Month 2012-07	0.018998	0.021238
M10	Month 2012-08	0.017871	0.020516

Model	Term	Coefficient	Clustered SE
M10	Month 2012-09	0.024207	0.021795
M10	Month 2012-10	0.001367	0.019895
M10	Month 2012-11	0.004110	0.020391
M10	Month 2012-12	0.015954	0.020005
M10	Month 2013-01	0.034949	0.019982
M10	Month 2013-02	0.019174	0.019885
M10	Month 2013-03	0.006973	0.019189
M10	Month 2013-04	0.020384	0.019713
M10	Month 2013-05	0.026149	0.019577
M10	Month 2013-06	0.007976	0.023195
M10	Month 2013-07	0.004817	0.031689
M10	Month 2013-08	0.020105	0.042341
M10	Month 2013-09	0.114018	0.027368
M10	Month 2013-10	0.002164	0.046281
M10	Month 2013-11	0.175185	0.071232
M10	Month 2013-12	0.036881	0.056719
M11	const	0.009838	0.005285
M11	buyer concession	-0.132780	0.028742
M11	experienced	-0.008804	0.003658
M11	interaction	0.035023	0.028596
M11	log buyer experience	0.000606	0.000408
M11	log initial seller price	-0.000040	0.000764
M11	gap before buyer concession	0.167032	0.009918
M11	Month 2012-07	0.000135	0.002423
M11	Month 2012-08	-0.000085	0.002298
M11	Month 2012-09	-0.001007	0.002427
M11	Month 2012-10	-0.000012	0.002565
M11	Month 2012-11	-0.000050	0.002367
M11	Month 2012-12	0.002529	0.002406
M11	Month 2013-01	0.000864	0.002365
M11	Month 2013-02	0.002473	0.002281
M11	Month 2013-03	0.002258	0.002307
M11	Month 2013-04	0.001706	0.002381
M11	Month 2013-05	0.003082	0.002437
M11	Month 2013-06	0.003017	0.003046
M11	Month 2013-07	0.002857	0.003598
M11	Month 2013-08	0.005261	0.004497
M11	Month 2013-09	0.012463	0.004386
M11	Month 2013-10	0.012644	0.005430
M11	Month 2013-11	0.015027	0.005222
M11	Month 2013-12	0.020675	0.007158

B Graph inputs and sample diagnostics

The following tables give every binned point, including the intermediate experience group not drawn in the main two-group curves. Intervals of buyer concessions are generated by right-closed 10 pp bins; zero is included in the first bin. Missing bins are omitted rather than assigned zero.

Table 9: Underlying raw-concession graph data

Experience	Buyer bin (%)	N	Mean buyer (%)	Mean seller (%)
0-9	0–10	50	3.920	25.039
0-9	10–20	109	16.682	22.855
0-9	20–30	141	25.515	24.596
0-9	30–40	159	35.815	28.642
0-9	40–50	168	47.653	28.288
0-9	50–60	56	55.533	24.626
0-9	60–70	41	65.326	30.185
0-9	70–80	14	75.080	28.201
0-9	80–90	4	83.698	20.833
10-99	0–10	257	3.737	18.171
10-99	10–20	492	16.300	21.741
10-99	20–30	599	25.687	27.066
10-99	30–40	826	35.782	26.965
10-99	40–50	774	47.473	26.067
10-99	50–60	249	55.313	29.550
10-99	60–70	132	65.299	23.986
10-99	70–80	62	74.500	26.408
10-99	80–90	11	85.812	46.234
10-99	90–100	1	99.667	100.000
100+	0–10	1,569	4.064	16.795
100+	10–20	3,512	16.292	20.583
100+	20–30	5,015	25.557	22.782
100+	30–40	6,299	35.422	25.332
100+	40–50	5,583	47.063	23.545
100+	50–60	2,574	54.840	25.585
100+	60–70	1,222	64.950	22.702
100+	70–80	420	73.961	22.486
100+	80–90	69	83.738	24.500
100+	90–100	11	94.656	36.364

Table 10: Underlying response-selection graph data

Experience	Buyer bin (%)	N	Mean buyer (%)	Accept (%)	Counter (%)
0-9	0–10	173	2.548	19.075	28.902
0-9	10–20	219	16.327	9.132	49.772
0-9	20–30	246	25.524	6.911	57.317
0-9	30–40	336	36.088	25.000	47.321
0-9	40–50	456	48.323	40.351	36.842
0-9	50–60	183	56.504	46.448	30.601
0-9	60–70	157	65.666	57.325	26.115

Experience	Buyer bin (%)	N	Mean buyer (%)	Accept (%)	Counter (%)
0-9	70–80	64	75.248	67.188	21.875
0-9	80–90	19	84.929	57.895	21.053
0-9	90–100	44	99.593	75.000	0.000
10-99	0–10	696	2.992	14.511	36.925
10-99	10–20	882	16.367	8.390	55.782
10-99	20–30	1,057	25.653	13.623	56.670
10-99	30–40	1,560	35.829	24.295	52.949
10-99	40–50	1,969	48.210	43.575	39.309
10-99	50–60	777	55.711	50.450	32.046
10-99	60–70	577	65.638	59.445	22.877
10-99	70–80	291	74.930	67.354	21.306
10-99	80–90	61	84.862	72.131	18.033
10-99	90–100	147	99.576	75.510	0.680
100+	0–10	4,128	3.292	13.614	38.009
100+	10–20	6,219	16.291	9.503	56.472
100+	20–30	8,390	25.577	15.721	59.774
100+	30–40	11,878	35.672	28.700	53.031
100+	40–50	14,420	47.781	47.767	38.717
100+	50–60	6,882	55.218	50.654	37.402
100+	60–70	4,729	65.348	64.749	25.841
100+	70–80	2,332	74.473	72.256	18.010
100+	80–90	542	84.243	77.306	12.731
100+	90–100	733	99.160	74.216	1.501

C Source checksum and licensing scope

The retained compressed byte range is 0–67,108,863. Its SHA-256 is

`c68403e3759c653ce5a1c5598a71b58d0d6ce`
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The source-reported compressed total is 1,374,076,192 bytes. The project retains complete CSV lines from the prefix and removes the final seller. The repository records separate hashes of the delivered processed files. The MIT license covers original code and report text, not the eBay/NBER data, source codebook or cited publications. Source-provider terms remain applicable.